

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Results-driven **AI Evaluation / Quality Engineering** professional with **6+ years of software engineering and test automation experience**, building production-grade validation, experimentation, and evaluation systems for enterprise and customer-facing applications.
- Hands-on experience applying **Evaluation-Driven Development and Experiment-Driven Development** to AI systems, using structured evaluation signals to guide prompt design, agent logic, model selection, system iteration, and production release decisions.
- Strong **Python engineering** expertise with experience developing clean, maintainable evaluation frameworks, experimentation pipelines, reusable testing utilities, scoring functions, regression infrastructure, and production automation services.
- Experienced building and maintaining **golden test cases, regression suites, domain-specific test scenarios, edge-case datasets, and AI-assisted test generation workflows** that evolve alongside AI systems as first-class system components.
- Hands-on experience designing **offline and online evaluation pipelines**, measuring system behavior against real user needs, domain constraints, business objectives, and production outcomes while avoiding overfitting to misleading metrics.
- Experienced defining and calibrating **LLM-based graders**, rubric-based scoring, automated quality signals, and human-in-the-loop evaluation by aligning machine judgments with expert and subject-matter-expert assessments.
- Strong systems-oriented and AI-native mindset, collaborating with AI Engineers, Architects, Product Engineers, AI Strategists, and domain experts to improve **prompts, agents, data, trust, safety, observability, and production reliability**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Engineering:** Python, pytest, Java, TypeScript, C#, SQL, REST APIs, JSON, Clean Code, Production Automation, Software Engineering
- **AI Evaluation:** Evaluation-Driven Development, Experiment-Driven Development, Evaluation Frameworks, Golden Test Cases, Regression Suites, Scoring Functions, Quality Signals, System Benchmarking
- **LLM & Agent Evaluation:** Generative AI, LLMs, Agentic AI, Prompt Evaluation, Agent Logic Validation, LLM-Based Graders, Rubric-Based Scoring, Model Selection, Release Readiness
- **Evaluation Pipelines:** Offline Evaluation, Online Evaluation, Production Evaluation Pipelines, Experimentation Pipelines, Human Evaluation, SME Calibration, Failure Analysis, Edge-Case Analysis
- **AI-Native Development:** AI-Assisted Test Generation, Synthetic Test Cases, Failure Analysis, Automated Debugging, Prompt Iteration, AI-Assisted Code Generation, Test Expansion
- **Systems & Production:** Agents, Prompts, Data Pipelines, REST APIs, Distributed Systems, CI/CD, Docker, AWS, Azure, Observability, Production Monitoring, Trust and Safety
- **Quality & Collaboration:** Human Judgment Encoding, Domain Constraints, Business Objectives, Customer-Facing AI, Regression Detection, Release Quality, Cross-Functional Collaboration, Technical Documentation

EXPERIENCE

Test Automation Engineer *PCMI LLC (Full Time)*

Oct 2024 – Present
San Francisco, CA, USA

- Designed and implemented production-grade **AI evaluation frameworks using Python**, applying Evaluation-Driven Development to measure system quality and use structured evaluation results to guide iterative improvements to prompts, agent logic, models, and application behavior.
- Built and maintained **golden test cases, regression suites, domain-specific scenarios, and edge-case datasets** using human-authored and AI-assisted test generation to capture critical customer behaviors and prevent regressions across AI system changes.
- Developed reusable **offline and online evaluation pipelines** integrated with system iteration and CI/CD workflows, enabling automated comparisons across prompt versions, agent configurations, model selections, and application releases.
- Defined domain-specific quality measurements aligned with real customer needs, operational constraints, and business objectives, avoiding reliance on generic metrics that did not accurately represent production outcomes.
- Designed and calibrated **LLM-based graders and rubric-based scoring functions**, comparing automated judgments against subject-matter-expert and human assessments to improve grader reliability and maintain high-quality evaluation signals.
- Used AI-native development practices to generate additional test scenarios, analyze failed evaluations, explore unexpected edge cases, accelerate root-cause analysis, and improve debugging and experimentation velocity.
- Analyzed divergence between offline evaluation scores and real-world production behavior, refining datasets, scoring logic, graders, and regression coverage to maintain trust and safety as AI systems evolved.
- Collaborated with AI Engineers, Architects, Product Engineers, domain experts, and business stakeholders to ensure evaluation frameworks meaningfully influenced prompt design, agent behavior, model selection, deployment decisions, and release readiness.

Environment: Python, pytest, Evaluation-Driven Development, Experiment-Driven Development, Generative AI, LLMs, Agentic AI, Golden Test Cases, Regression Suites, Offline Evaluation, Online Evaluation, LLM-Based Graders, Rubric-Based Scoring, AI-Assisted Test Generation, Prompt Evaluation, Agent Logic, Model Selection, REST APIs, Docker, AWS, CI/CD, Production Monitoring, Trust and Safety

SDET

Dec 2023 – Sept 2024
Remote, USA

Client: CVS

- Developed Python-based evaluation and experimentation frameworks for **customer-facing AI and Generative AI healthcare applications**, using measurable quality signals to drive system changes and release-readiness decisions.
- Created and continuously evolved **golden datasets, regression test suites, representative healthcare workflows, and difficult edge cases** covering customer journeys, data variations, model behavior, and production-risk scenarios.
- Built offline evaluation pipelines to benchmark prompt, model, retrieval, and agent changes before release, and online validation workflows to compare production behavior against expected quality outcomes.
- Worked with Product, Engineering, and healthcare subject-matter experts to translate qualitative human judgments into structured **test cases, scoring functions, evaluation rubrics, and automated graders**.
- Implemented and calibrated LLM-based graders, validating automated judgments against expert human review and investigating disagreement patterns to improve grading accuracy and signal quality.
- Applied AI-assisted test generation to expand evaluation coverage, discover failure modes, identify underrepresented scenarios, analyze regressions, and accelerate investigation of production-quality issues.
- Investigated evaluation failures across prompts, agents, APIs, data, and system integrations, determining whether issues resulted from model behavior, test-data gaps, grader limitations, or application logic.
- Partnered with AI engineers, Product Owners, domain experts, and platform teams to use evaluation findings to improve system reliability, customer outcomes, deployment confidence, and production trust.

Environment: Python, pytest, Generative AI, LLMs, Evaluation Frameworks, Golden Datasets, Regression Suites, LLM-Based Graders, Human Evaluation, SME Calibration, Offline Evaluation, Online Evaluation, AI-Assisted Test Generation, Prompt Testing, Agent Evaluation, Healthcare Workflows, REST APIs, AWS, Docker, CI/CD, Jira, Production Validation

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Developed clean and reusable Python and Java automation frameworks for enterprise applications, creating structured testing and experimentation pipelines that provided repeatable quality signals throughout development.
- Designed regression suites, reusable datasets, scoring logic, and automated validations to compare application behavior across software revisions and identify changes that negatively affected customer workflows.
- Built data-driven evaluation utilities to convert business requirements and domain expectations into measurable test conditions, assertions, scoring criteria, and repeatable acceptance checks.
- Developed automated pipelines for API, integration, data, regression, and end-to-end testing, embedding validation directly into CI/CD workflows to provide rapid feedback during application iteration.
- Analyzed test and production failures using Python, application logs, API responses, SQL data, and system telemetry, distinguishing application defects from test-data, environment, and automation-framework issues.
- Worked closely with Product Managers, Software Engineers, and domain stakeholders to translate human expectations into explicit test cases, validation rules, and measurable quality criteria.
- Improved regression coverage by identifying recurring failure modes and edge cases, expanding test datasets and automation scenarios to protect critical behaviors across subsequent releases.
- Produced clear technical reports and quality summaries that connected test results with business impact, supporting evidence-based decisions on software changes and production readiness.

Environment: Python, Java, pytest, TestNG, REST APIs, SQL, Data-Driven Testing, Regression Suites, Evaluation Metrics, Automated Validation, Experimentation, CI/CD, Jenkins, Git, Docker, Failure Analysis, Quality Signals, Technical Reporting, Agile Scrum

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed reusable automation frameworks using Python, Java, Selenium WebDriver, Cucumber BDD, Appium, and TestNG to systematically evaluate enterprise web, mobile, API, and backend application behavior.
- Built and maintained regression suites containing representative workflows, boundary scenarios, negative tests, and edge cases to ensure critical system behavior remained stable across application changes.
- Translated business requirements and stakeholder expectations into structured test cases, automated assertions, measurable pass/fail criteria, and repeatable validation workflows.
- Created reusable test datasets and data-driven test scenarios to increase behavioral coverage and evaluate application responses across varying inputs, configurations, and customer scenarios.
- Investigated failed validations using application logs, API outputs, SQL queries, screenshots, and execution traces, identifying defects and distinguishing product failures from environment and test-automation issues.
- Integrated automated regression suites into Jenkins, Git, Maven, and CI/CD workflows to ensure evaluation and validation occurred continuously as software evolved.
- Collaborated with developers, Business Analysts, QA leads, and Product stakeholders to refine test scenarios, improve quality measurements, expand coverage, and validate high-priority customer workflows.
- Prepared structured defect reports, test summaries, regression findings, and release-readiness recommendations, providing stakeholders with evidence-based quality signals for deployment decisions.

Environment: Python, Java, Selenium WebDriver, Appium, Cucumber BDD, TestNG, REST APIs, SQL, Regression Suites, Data-Driven Testing, Automated Evaluation, Edge-Case Testing, Jenkins, Git, Maven, CI/CD, Failure Analysis, Quality Metrics, Technical Documentation, Agile Scrum

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.